

# 2+1 Dimensional Gravity and Chern-Simons Theory

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## Abstract

In this note, I'll start to learn about the Chern-Simons formulation of 2+1 dimensional gravity, following the paper of Witten (1988)[1] . For I'm reading this paper while lacking the knowledge of Basic Chern-Simons Theory and Classical 2+1 Dim Gravity, I'll include some background materials as well as preliminaries.

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# 1 Preliminaries I: Topology and Geometry

Here I will give a brief review of some basic concepts in topology that will be useful in understanding the Chern-Simons formulation of 2+1 dimensional gravity. It mainly follows the treatment in [2] However, it is not enough or rigorous for understanding any steps in Witten's paper, but serve as an introduction to many nouns he used in the paper.

## 1.1 Homeomorphisms and diffeomorphisms

Consider two topological manifold  $X$  and  $Y$ . We can define a **homeomorphism** between them:

**Definition 1.** A *homeomorphism* is a continuous bijection  $f : X \rightarrow Y$  with a continuous inverse  $f^{-1} : Y \rightarrow X$ .

If such a homeomorphism exists, we say that  $X$  and  $Y$  are **homeomorphic**, denoted as  $X \cong Y$ . We view the two manifold as "topologically equivalent", meaning they have the same topological properties.

If the manifolds are differential manifolds, we can further define a **diffeomorphism**:

**Definition 2.** A *diffeomorphism* is a smooth bijection  $f : X \rightarrow Y$  with a smooth inverse  $f^{-1} : Y \rightarrow X$ .

If we focus on 3 dimensional manifolds, there is an important theorem:

**Theorem 1.** *Moise's Theorem*

*For 2 & 3 dimensional manifolds, being homeomorphic is equivalent to being diffeomorphic.*

This theorem implies that in 2 and 3 dimensions, the topological structure and the smooth structure are essentially the same. This is not true in higher dimensions, where there exist manifolds that are homeomorphic but not diffeomorphic (exotic spheres).

## 1.2 Mapping Class Group

### 1.2.1 Isotopy

We focus on the self-diffeo of one manifold  $\mathcal{M}$ ,  $f : \mathcal{M} \rightarrow \mathcal{M}$ . We can define an **Isotopy** between two diffeomorphisms:

**Definition 3.** An *isotopy* between two diffeomorphisms  $f, g : \mathcal{M} \rightarrow \mathcal{M}$  is a continuous family of diffeomorphisms  $F(s) : \mathcal{M} \rightarrow \mathcal{M}$  for  $s \in [0, 1]$  such that  $F(0) = f$  and  $F(1) = g$ . If an Isotopy exists between  $f$  and  $g$ , we say that they are *isotopic*.

Most diffeos considered in physics are isotopic to identity map, yet there for topologically non-trivial manifolds, there can be diffeos that are not isotopic to identity.

- eg. **Dehn Twist** on torus  $T^2$ : Cut the torus along a non-contractible loop, twist one side by  $360^\circ$ , and then re-glue. This operation cannot be continuously deformed to the identity map.

### 1.2.2 Mapping Class Group

We can define an equivalence relation using Isotopy and define its equivalent class on the set of self-diffeos of  $\mathcal{M}$ :

**Definition 4.** The **Mapping Class Group** (MCG) of a manifold  $\mathcal{M}$ , denoted as  $MCG(\mathcal{M})$ , is the group of isotopy classes of self-diffeomorphisms of  $\mathcal{M}$ .

We can prove that the equivalent class set has a group structure.

The mapping class group of 2 dimensional surfaces is well studied and they are generated by:

- **Dehn Twists** along non-contractible loops.

## 1.3 Fundamental Group

### 1.3.1 Definition

**Definition 5.** *Homotopy Curve*

Consider a series of curves satisfying:  $\gamma : [0, 1] \rightarrow M$ ,  $\gamma(0) = x_0$ ,  $\gamma(1) = x_0$ . If two curves  $\gamma_0, \gamma_1$  can be continuously deformed to each other while keeping the endpoints fixed, we say they are **homotopic**, denoted as  $\gamma_0 \simeq \gamma_1$ .

Mathematically, there exists a continuous map  $F : [0, 1] \times [0, 1] \rightarrow M$  such that:

$$\begin{aligned} F(t, 0) &= \gamma_1(t) \\ F(t, 1) &= \gamma_2(t) \\ F(0, s) &= F(1, s) = x_0. \end{aligned} \tag{1}$$

**Definition 6.** *Fundamental Group*

For a fixed base point  $x_0$  in a topological space  $M$ , the **fundamental group**  $\pi_1(M, x_0)$  is the set of equivalence classes of loops  $[\gamma]$  based at  $x_0$  under the homotopy relation.

We can prove that the set of equivalence classes has a group structure, with the group operation defined as the concatenation of loops.

**Theorem 2.** For connected manifolds, the fundamental group is independent of the choice of base point up to isomorphism. Therefore, we often denote it simply as  $\pi_1(M)$ .

### 1.3.2 More on Properties

**Definition 7.** *Simply Connected*

A topological space  $M$  is said to be **simply connected** if it is path-connected and its fundamental group is trivial, i.e.,  $\pi_1(M) = \{e\}$ , where  $e$  is the identity element.

**Theorem 3.** *Topological Invariance of Fundamental Group*

If two topological spaces  $M$  and  $N$  are homeomorphic, then their fundamental groups are isomorphic, i.e.,  $\pi_1(M) \cong \pi_1(N)$ .

**Theorem 4.** *Fully Characterization of 2D Surfaces by Fundamental Group*

*The fundamental group fully characterizes the topology of closed 2-dimensional surfaces. Two closed surfaces are homeomorphic if and only if their fundamental groups are isomorphic.*

## 1.4 Covering Space

### 1.4.1 Covering Space Definition

We can unwrap the non-contractible loops of a manifold by considering its covering space.

**Definition 8.** *Covering Space*

A **covering space** of a topological space  $M$  is a topological space  $\tilde{M}$  together with a continuous surjective map  $p : \tilde{M} \rightarrow M$  such that

- For every point  $x \in M$ , there exists an open neighborhood  $U$  of  $x$  such that the preimage  $p^{-1}(U)$  is homeomorphic to  $U \times \Lambda$ , where  $\Lambda$  is a discrete set.

### 1.4.2 Topology of Covering Space

The surjective map used by defining the covering space can induce a homomorphism between the fundamental groups of the two spaces:

**Theorem 5.** *Induced Homomorphism on Fundamental Groups*

Let  $p : \tilde{M} \rightarrow M$  be a covering map. Then, for any base point  $\tilde{x}_0 \in \tilde{M}$  with  $p(\tilde{x}_0) = x_0 \in M$ , the map  $p$  induces a homomorphism  $p_* : \pi_1(\tilde{M}, \tilde{x}_0) \rightarrow \pi_1(M, x_0)$ .

This homomorphism is injective but not necessarily surjective. The image of  $p_*$  is a subgroup of  $\pi_1(M, x_0)$  called the **characteristic subgroup** of the covering space. It only consists of paths that are lifted to  $\tilde{M}$  to be loops instead of open curves.

**Theorem 6.** *For a connected manifold, any subgroup of its fundamental group can be realized as the characteristic subgroup of some covering space.*

Thus we can classify the covering spaces with the subgroups of the fundamental group.

**Definition 9.** *Universal Covering Space*

A **universal covering space** of a topological space  $M$  is a covering space  $\tilde{M}$  that is simply connected, i.e., its fundamental group is trivial:  $\pi_1(\tilde{M}) = \{e\}$ . which corresponds to the trivial subgroup of  $\pi_1(M)$ .

## 1.5 Quotient Space

### 1.5.1 Quotient Space Definition

We can construct a new manifold by quotienting a manifold with a group action.

**Definition 10.** *Quotient Space*

Given a group  $G$  acting on a manifold  $M$ . The orbit of a point  $x \in M$  under the action of  $G$  is the set  $G \cdot x = \{g \cdot x | g \in G\}$ . The **quotient space**  $M/G$  is the set of all orbits of points in  $M$  under the action of  $G$ , equipped with the quotient topology.

We can define a natural projection from the points in  $M$  to their corresponding orbits in  $M/G$ , which gives a quotient topology by defining open sets in  $M/G$  as those whose preimages under the projection are open in  $M$ .

- In general a Quotient Space **May NOT be a Manifold**.

**Theorem 7.** *Quotient Space as Manifold*

*If a group  $G$  acts properly discontinuously and freely on a manifold  $M$ , then the quotient space  $M/G$  is also a manifold.*

- **freely:** For any point  $x \in M$ , if  $g \cdot x = x$  for some  $g \in G$ , then  $g$  must be the identity element of  $G$ . Equivalently the isotropy group  $G_x = \{g \in G | g \cdot x = x\}$  is trivial for all  $x \in M$ .
- **Properly Discontinuously:**
  - Any two points in different orbits have neighborhoods that do not intersect under the group action.
  - for any point  $x$  the isotropy group  $G_x = \{g \in G | g \cdot x = x\}$  is finite.
  - For any point  $x \in M$ , there exists an open neighborhood  $U$  of  $x$  such that for all  $g \in G$  not in the isotropy group  $G_x$ , the images  $g \cdot U$  do not intersect with  $U$ , i.e.,  $g \cdot U \cap U = \emptyset$ . for all  $g \in G_x$ , we have  $g \cdot U = U$ .

**Definition 11.** *Orbifold*

*If a group  $G$  acts properly discontinuously but not freely on a manifold  $M$ , the quotient space  $M/G$  is called an **orbifold**.*

An orbifold is a generalization of manifold with some mild singularities.

We can find a point on  $M$  to label the orbits in  $M/G$ , the complete set of such points is called a **fundamental region** of the group action.

### 1.5.2 Topology of Quotient Space

**Theorem 8.** *Topology of Quotient Space of Simply Connected Manifold*

*If  $M$  is a simply connected manifold and  $G$  is a finite group then we can prove a relation between the fundamental group of the quotient space and the group action:*

$$\pi_1(M/G) \cong G. \quad (2)$$

In this case the projection  $p : M \rightarrow M/G$  is the universal covering map of the quotient space and  $M$  is the universal covering space of  $M/G$ .

**Theorem 9.** *Connected Manifold as Quotient Space*

*Any connected manifold  $M$  can be represented as the quotient space of its universal covering space  $\tilde{M}$  by the action of its fundamental group  $\pi_1(M)$ :*

$$M \cong \tilde{M} / \pi_1(M). \quad (3)$$

*Here we define the action of  $\pi_1(M)$  on  $\tilde{M}$  as: Mapping the loop in  $M$  mapped to open path in  $\tilde{M}$  back by identifying its endpoints.*

Mathematically, this action of  $\pi_1(M)$  on  $\tilde{M}$  is defined as a **Deck Transformation**:

**Definition 12.** *Deck Transformation*

For any  $[\gamma] \in \pi_1(M)$  we can define a map  $D_{[\gamma]} : \tilde{M} \rightarrow \tilde{M}$ , that satisfy  $p \circ D_{[\gamma]} = p$ . It gives a representation of  $\pi_1(M)$  as a group of homeomorphisms of  $\tilde{M}$ .

## 1.6 Geometization in 2D

Before we only consider topological manifolds, now we can further consider differential manifolds with riemannian metrics.

### 1.6.1 Standard Geometries in 2D

In 2 dimensions, there are three standard geometries with constant curvature:

- **Riemann Sphere**  $\hat{\mathbb{C}} = \mathbb{C} \cup \infty$  with constant positive curvature metric:

$$ds^2 = \frac{4dzd\bar{z}}{(1 + |z|^2)^2}. \quad (4)$$

- **Complex Plane**  $\mathbb{C}$  with flat metric:

$$ds^2 = dzd\bar{z}. \quad (5)$$

- **Hyperbolic Plane**  $\mathbb{H}^2 = \{z \in \mathbb{C} | \text{Im}(z) > 0\}$  with constant negative curvature metric:

$$ds^2 = \frac{dzd\bar{z}}{(\text{Im}(z))^2}. \quad (6)$$

### 1.6.2 Uniformization Theorem

We give out one of the most important theorems in 2D geometry:

**Theorem 10.** *Uniformization Theorem(Pure Topology Version)*

Every Surface is homeomorphic (diffeomorphic) to  $D/\Gamma$ , where  $D \in \{\hat{\mathbb{C}}, \mathbb{C}, \mathbb{H}^2\}$  and  $\Gamma \subset \text{Iso}(D)$  is a properly discontinuously isometry on  $D$ .

{thm:Unif

This is a topological statement, we only say that a topological surface can be represented as a quotient space of one of the three standard geometries. However, with suitable operations we can give the quotient space a constant curvature metric induced from the standard geometry. [It is infact quite important to know this procedure though it seems to be doable] Thus, we know that:

- Any closed surface can be given a constant curvature metric.
- Classification of such metric = Classification of isometry group actions on the standard geometries.

## 2 Preliminaries II: 1st Order Formalism of Gravity

I mainly follow the discussion in [2]. For more detailed discussion and having the same notation one can refer to [3].

### 2.1 Vielbein and Spin Connections

We can a GR theory using a first order formalism, which use frame (aka vielbein) and spin connection as basic variables instead of metric. This is especially useful in 2+1 dimensional gravity, since it can be reformulated as a Chern-Simons theory. We define the frame fields

**Definition 13.** *Frame Fields are a set of orthogonal vectors in the tangent space  $e^\mu_a$  or the inverse one form  $e_\mu^a$ , which satisfy:*

$$g^{\mu\nu} e_\mu^a e_\nu^b = \eta^{ab}, \quad e_\mu^a e_\nu^b \eta_{ab} = g_{\mu\nu}. \quad (7)$$

Here  $e^\mu_a$  is the inverse of  $e_\mu^a$ , i.e.,

$$e^\mu_a e_\mu^b = \delta_a^b, \quad e_\mu^a e^\nu_a = \delta_\mu^\nu. \quad (8)$$

We can show that  $e^a = e_\mu^a dx^\mu$  forms a orthonormal basis of the cotangent space and  $e_a = e^\mu_a \partial_\mu$  forms a orthonormal basis of the tangent space. The coefficients can also be understood as transformation matrices between coordinate basis and orthonormal basis.

This set of vectors definately forms a basis of the tangent space, thus we can consider the connections in this basis. For example, the Christoffel symbols can be written as:

$$\nabla_\mu \partial_\nu = \Gamma_{\mu\nu}^\rho \partial_\rho, \quad \text{or} \quad \nabla_\mu dx^\nu = -\Gamma_{\mu\rho}^\nu dx^\rho \quad (9)$$

Thus, a analogue is to define a connection  $\omega_\mu^a{}_b$  as

**Definition 14.** *Spin Connection  $\omega_\mu^a{}_b$  is defined as:*

$$\nabla_\mu e_\nu^a = -\omega_\mu^a{}_b e_\nu^b \quad (10)$$

#### Remark:

The covariant derivative here  $\nabla_\mu e_\nu^a$  means that we only view  $e_\nu^a$  as a dual vector in the cotangent space, with componet  $\mu$  of the coordinate basis. Sometimes you may encounter the notation:

$$\nabla_\mu e_\nu^a = \partial_\mu e_\nu^a - \Gamma_{\mu\nu}^\lambda e_\lambda^a + \omega_\mu^a{}_b e_\nu^b \quad (11)$$

This is when we view it as a (1,1) tensor in a mixed basis, with the coordinate basis in the cotangent space and the orthonormal basis in the tangent space.

The covariant derivative of a vector is a rank (0,2) tensor, thus we know that  $\omega_\mu^a{}_b$  transforms as a one form in the  $\mu$  index. Thus:

- Both  $e^a = e_\mu^a dx^\mu$  and  $\omega^a{}_b = \omega_\mu^a{}_b dx^\mu$  can be understood as one forms. The inverse of the frame fields can be written as a vector field  $e_a = e^\mu_a \partial_\mu$ .
- We define the lowering and raising of the spin connection indices  $\omega_{ab} = \eta_{ac} \omega^c{}_b$ .



**Remark:**

Notice that in fact we can lower and raise the  $\mu, \nu$  indices with  $g$  and raise and lower the  $a, b$  indices with  $\eta$ . This is totally consistent with the definition of frame fields.

We can prove that if the theory is metric compatible, then the spin connection 1-form is antisymmetric in  $a, b$  indices:

$$\omega_{ab} = -\omega_{ba}. \quad (12)$$

## 2.2 Cartan Structure Equations

Using the above definitions, we hope to express important geometric quantities such as Riemann curvature Tensor. Moreover, we also notice that the frame fields and the spin connections are not independent, just like the Christoffel symbol can be represented by the metric. We want to:

- Represent the spin connection in terms of the frame fields or at least find a relation between them.
- Represent the Riemann curvature Tensor in terms of the spin connection.

These two goals can be achieved by the Cartan structure equations:

**Theorem 11.** *Cartan's first structure equation*

*If the theory is torsion free, then the frame fields and the spin connection satisfy:*

$$D_\omega e^a \equiv de^a + \omega^a_b \wedge e^b = 0, \quad (13)$$

$D_\omega e^a$  is called the gauge covariant derivative of  $e^a$  with respect to the spin connection  $\omega$ .

This can be proved directly from the definition of the covariant derivative and the definition of the spin connection. The torsion free condition makes the lower indices of the Christoffel symbol symmetric, thus it cancels out when doing an exterior derivative.

**Remark:**

Note that exterior derivative is using the operator  $d$  which is defined using the derivative  $\partial$  not  $\nabla$ .

Then where is the Riemann curvature tensor? It can be found in the second structure equation:

**Theorem 12.** *Cartan's second structure equation*

*The Riemann curvature 2-form can be written as:*

$$R^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b. \quad (14)$$

Here the Riemann curvature 2-form is defined as:

$$R^a_b = R^a_{b\mu\nu} dx^\mu \wedge dx^\nu \quad (15)$$

## 2.3 Local Lorentz Transformations

[This is important see Carroll]

Using the vielbein formalism, we can have more redundancy in describing the spacetime. This is because at each point we can choose different orthonormal basis in the tangent space, which are related by local Lorentz transformations.

If we use the notation of  $\partial_\mu = e_\mu^a e_a$  and  $e_a = e_a^\mu \partial_\mu$ , then a local Lorentz transformation can be written as:

### 3 Preliminary III: 2+1 Dimentional Classical Gravity

{sec:}

Here I'll talk about a standard set up in studying 2+1 dimensional gravity.

#### 3.1 Local Structure

The Riemannian curvature of 2+1 dimensional gravity can be written directly in terms of Ricci Tensor and the metric:

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} + g_{\nu\sigma}R_{\mu\rho} - g_{\nu\rho}R_{\mu\sigma} - g_{\mu\sigma}R_{\nu\rho} - \frac{1}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})R \quad (16)$$

Thus if we consider the vaccum Einstein equation  $R_{\mu\nu} = 0$ , the Riemannian curvature vanishes identically! And if we consider the cosmological constant case  $R_{\mu\nu} = \Lambda g_{\mu\nu}$ , the Riemannian curvature becomes:

$$R_{\mu\nu\rho\sigma} = \Lambda(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}) \quad (17)$$

This is a Riemann Tensor of a constant convature spacetime.

- When  $\Lambda = 0$ , the spacetime is locally flat, i.e., locally isometric to Minkowski space  $\mathbb{R}^{2,1}$ .
- When  $\Lambda > 0$ , the spacetime is locally de Sitter space  $dS_3$ .
- When  $\Lambda < 0$ , the spacetime is locally anti-de Sitter space  $AdS_3$ .

#### 3.2 Global Structure

However, being locally constant curvature spacetime doesn't mean that the theory is trivial, it may include various global properties. However, we shall ask what kinds of global topology may admit a constant curvature metric as well as a lorentzian structure(for physical spacetime is lorentzian).

The answer is given by the following theorem:

**Theorem 13.** *Let  $M$  be a compact 3 manifold with a flat, time-orientable Lorentzian metric and a purely spacelike boundary. Then  $M$  has the topology of:*

$$M \cong [0, 1] \times \Sigma \quad (18)$$

*Here  $\Sigma$  is a closed surface(compact and no boundary) homeomorphic to one of the boundary components of  $M$ .*

Thus the topology of a physical spacetime is fixed by the topology of a initial spacelike slice  $\Sigma$ . In the following context we shall only focus on this kind of spacetime topology.

#### Remark:

In Witten's paper, he seem to not distinguish the topology of  $[0, 1] \times \Sigma$  and  $\mathbb{R} \times \Sigma$ . I won't be distinguishing it further, since it seems not causing any physical consequence in the phase space discussion.

#### 3.3 Geometric Phase Space Analysis

Here we follow chapter 4 of Carlip's book [2] to do a geometric phase space analysis of 2+1 dimensional pure gravity.

## 4 Chern-Simons Theory and Gravity (Witten)

Here we continue on following Witten's discussion on the relation between 2+1 dimensional gravity and Chern-Simons theory [1].

### 4.1 Geometric Phase Space Analysis

[Plan to not keep this , its unclear!!]

#### 4.1.1 Discussion Set Up

How to quantize a Theory, we usually start from analyzing the phase space of the theory and then construct a poisson braket of the phase space. Then one can transform the poisson bracket into commutator and get the quantum theory. Thus to study the quantum gravity in 2+1 dimensions, we need to first learn about the phase space of the classical 2+1 dimensional gravity.

The phase space of a classical theory is often defined as:

**Definition 15.** *Phase Space*

*The phase space is the space of all  $q, \dot{q}$  at  $t = 0$  surface.*

However, this definition is not covariant and we need tons of constraints to make it work, thus in 2+1 dimensional gravity, we use the following definition:

**Definition 16.** *Phase Space*

*The phase space is the space of all classical solutions to the equations of motion Modulo gauge transformations.*

For 2+1 dimensional gravity, this means that we want to find all the solutions to:

$$R_{\mu\nu} = 0 \tag{19}$$

To do this, we might first assume a spacetime topology of the manifold we want to consider, we focus on a manifold:

- With topology  $M \cong [0, 1] \times \Sigma$ . where  $\Sigma$  is a genus  $g$  closed surface.

We now simplify the problem of analyzing the phase space of 2+1 dimensional gravity to finding the space of all solutions to the vacuum Einstein equation  $R_{\mu\nu} = 0$  on a manifold with topology  $M \cong [0, 1] \times \Sigma$ . These solutions can be found by quotienting the Minkowski space by a discrete subgroup of the isometry group of Minkowski space, which is the Poincare group  $ISO(2, 1)$ . In Witten's paper he gives out a analysis of those manifolds.

#### 4.1.2 Manifolds with initial & final singularities

Witten explicitly constructs these class of solutions as examples, by quotienting the light cone of Minkowski, now we explain the procedure:

- **Preparation: Quotienting  $\mathbb{H}$  to  $\Sigma_g$**

We can construct a 2-manifold with genus  $g \geq 2$  by quotienting the hyperbolic plane.

- **Step 1:** We first find the fundamental group of the surface  $\Sigma_g$  which is  $\pi_1(\Sigma_g)$ .
- **Step 2:** We find a subgroup of  $PSL(2, \mathbb{R})$  naming  $\Gamma$  which is isomorphic to  $\pi_1(\Sigma_g)$  and acts **discretely** on  $\mathbb{H}$ .

- **Step 3:** The quotient  $\mathbb{H}/\Gamma$  defines a Riemann surface with genus  $g$  and a metric with constant negative curvature.

[In fact I don't understand how this step works mathematically, eg how to construct the metric and why we don't require the group to be free but only properly continuous (discrete), etc. I need to learn more about this step.]

**Remark:**

Note that here we quotient by a discrete subgroup other than a properly discontinuous subgroup, because the action of a properly discontinuous subgroup is equivalent to being discrete in the case of being a subgroup of  $Aut(\mathbb{H}) = PSL(2, \mathbb{R})$ .

Then we apply a similar procedure to the light cone of Minkowski space to construct a flat 3-manifold with topology  $M \cong [0, 1] \times \Sigma_g$  with initial singularity and final singularity.

To do this we first list out some useful facts about the 2+1 Minkowski spacetime:

- Normally we have the metric of:

$$ds^2 = -dt^2 + dx^2 + dy^2 \quad (20)$$

And the light cone is defined by:

- $X^+$  future light cone:  $t^2 > x^2 + y^2, t > 0$
- $X^-$  past light cone:  $t^2 > x^2 + y^2, t < 0$
- The isometry group is the 3 dimensional Poincare group  $ISO(2, 1)$  which is the semidirect product of the Lorentz group  $SO(2, 1)$  and the translation group  $\mathbb{R}^{2,1}$ .
- The proper orthochronous Lorentz group  $SO^+(2, 1)$  is isomorphic to  $PSL(2, \mathbb{R})$ .

**Quotienting the light cone to get a 3-manifold with initial & final singularities**

We first notice that the light cone can be made up by hyperbolic planes:

$$t^2 = x^2 + y^2 + \tau^2 \quad (21)$$

where  $\tau$  is a parameter for the hyperbolic plane.

- Each hyperbolic plane is isometric to  $\mathbb{H}$  and the isometry group of the hyperbolic plane is the Lorentz group  $SO^+(2, 1)$  which is a subgroup of the Poincare group  $ISO(2, 1)$

Thus given a  $\Sigma_g$  we can construct a subgroup of the lorentz group  $\Gamma$  isometric to  $\pi_1(\Sigma_g)$  and quotient the hyperbolic plane by this subgroup to get a  $\Sigma_g$ . If we do this quotient for the whole  $X^+$  light cone we can get a 3 manifold with topology  $M \cong [0, 1] \times \Sigma_g$  and with an initial singularity at  $\tau = 0$ .

- The resulting manifold is a solution to the vacuum Einstein Field Equation for it is constructed by quotienting an isometry of Minkowski spacetime. This quotient group preserves the metric thus the resulting manifold is still flat.

[I don't understand why?? It needs knowledge of metric under quotienting.]

- The resulting manifold has a  $\Sigma_g \times [0, 1]$  topology. This is because the lorentz group act the hyperbolic plane to itself due to its feature of preserving Minkowski metric. Thus,  $\Gamma$  as a subgroup of lorentz group act independently on each hyperbolic plane.

Moreover, the isometry group of the hyperbolic plane is the lorentz group which is isomorphic to  $PSL(2, \mathbb{R})$  and the hyperbolic plane is isometric to  $\mathbb{H}$ . Thus, quotienting the hyperbolic plane with the subgroup  $\Gamma$  of the lorentz group is equivalent to quotienting  $\mathbb{H}$  with the same subgroup, which gives us a  $\Sigma_g$ . Thus, the resulting manifold has a  $\Sigma_g \times [0, 1]$  topology.

### Parameters Counting

Finally, we can count how many distinct manifold can be constructed by this procedure, the answer depends on the choice of  $\Gamma$  subgroup. The final answer is that the moduli of genus  $g$  surface, which is  $\mathbb{R}^{6g-6}$  gives us the label of distinct manifolds we can construct by this procedure.

[I don't understand this counting, it need some knowledge of Teichmuller space and moduli space of Riemann surface.]

#### 4.1.3 General Flat Manifolds

Now we try to construct more general flat manifolds with topology  $M \cong [0, 1] \times \Sigma_g$  by quotienting more general subspace of the Minkowski space.

For a general flat 3-manifold  $\hat{M}$ , we can construct its simple connected universal cover  $M$ . Due to the flatness and being a universal covering space,  $M$  is isometric to the Minkowski space or a subspace of the Minkowski space. And by definition,  $\hat{M}$  must be a quotient of  $M$ , which is a subspace of the Minkowski space, by a subgroup of the isometry group of the Minkowski space, which is the Poincare group  $ISO(2, 1)$ .

Under this set up, we consider the fundamental group of  $\hat{M}$ , which is  $\pi_1(\hat{M})$ . Every element shall be a loop in  $\hat{M}$ , which can be lifted to a path in  $M$ . Then we can define a element of the Poincare group  $ISO(2, 1)$   $\phi(\gamma)$  which ensures that the path  $[\gamma]$  in  $M$  which lifted from the loop  $[\gamma]$  in  $\hat{M}$ , is mapped to a noncontractible loop after the quotienting by  $\phi(\gamma)$ .

Thus we can see that:

- Every flat 3-manifold give rise to a homomorphism from  $\pi_1(\hat{M})$  to the Poincare group  $ISO(2, 1)$ . It can be constructed by quotienting Minkowski space by the image of the homomorphism.

[but why this procedure can also cover quotienting a subspace of Minkowski space??? It only covers the case of quotienting the whole Minkowski space, but not the case of quotienting a subspace of Minkowski space, which is also a solution to the Einstein equation. I think perhaps this is one of the non-rigorous step in this derivation. Thus, we need a more rigorous field theoretical proof of the phase space.]

Thus, the phase space of 2+1 dimensional gravity can be identified with the following space:

$$\text{Phase Space} = \text{Hom}(\pi_1(\Sigma_g), ISO(2, 1)) / \sim \quad (22)$$

Here  $\sim$  means that we quotient by the conjugation of the Poincare group  $ISO(2, 1)$ . We know that if two homomorphisms  $\phi_1, \phi_2$  are related as:

$$\phi_1(\gamma) = g\phi_2(\gamma)g^{-1} \quad \forall \gamma \in ISO(2, 1) \quad (23)$$

Then they give rise to the same manifold under quotient.

If we want to consider the topology of  $\hat{M} = \Sigma_g \times [0, 1]$ , then the fundamental group of the manifold is  $\pi_1(\hat{M}) = \pi_1(\Sigma_g)$ . Thus, the phase space of 2+1 dimensional gravity with topology  $M \cong [0, 1] \times \Sigma_g$  is:

**Theorem 14.** *The phase space of 2+1 dimensional gravity with topology  $M \cong [0, 1] \times \Sigma_g$  is:*

$$\text{Phase Space} = \text{Hom}(\pi_1(\Sigma_g), ISO(2, 1)) / \sim \quad (24)$$

where  $\sim$  means that we quotient by the conjugation of the Poincare group  $ISO(2, 1)$ .

### Parameter Counting

We then count the dimension of the phase space. For a surface  $\Sigma_g$ , its fundamental group can be generated by  $2g$  generators  $a_i, b_i$  with one relation:

$$\pi_1(\Sigma_g) = \langle a_1, b_1, \dots, a_g, b_g | \prod_{i=1}^g a_i b_i a_i^{-1} b_i^{-1} = 1 \rangle \quad (25)$$

Thus a homomorphism from  $\pi_1(\Sigma_g)$  to the Poincare group  $ISO(2, 1)$  is determined by the image of the generators  $a_i, b_i$  in the Poincare group  $ISO(2, 1)$ . Which lead to  $2g - 1$  parameters with each parameter the dimension of the Poincare group  $ISO(2, 1)$ , which is 6. Moreover, we need to quotient the phase space by the conjugation of the Poincare group  $ISO(2, 1)$ , which lead to a reduction of 6 parameters. Thus, the dimension of the phase space is:

$$\dim(\text{Phase Space}) = (2g - 1) \cdot 6 - 6 = 12g - 12 \quad (26)$$

**Remark:**

Finally, we shall note that this investigation is not complete and rigorous and as Witten have noted that we need a field theoretical proof of the phase space. Moreover, though we can construct the solutions, we also have left the problem that the quotienting may lead to manifold with undesirable properties, though being flat.

In the final section section 5 we will view the phase space from a more rigorous CS field theory approach.

## 4.2 Action of Vierbein Formalism

The Vielbein formalism of gravity can be written from a action principle.

## 4.3 Review on Chern-Simons Theory

## 4.4 3D Gravity Action as Chern-Simons Theory

## 4.5 On-Shell Diffeomorphism as Gauge Transformation

## 5 Quantization and Phase Space Analysis (Witten)

{sec:Quantization}

This is quantization!!

## 6 AdS3 Phase Space

Here we follow the discussion in [4] to give an illustration of the modern perspective of the gravitational phase space of AdS3.

{sec:Phas



## References

- <sup>1</sup>E. Witten, “2 + 1 dimensional gravity as an exactly soluble system”, Nuclear Physics B **311**, 46–78 (1988).
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- <sup>3</sup>S. M. Carroll, *Spacetime and geometry: an introduction to general relativity*, Pearson new international edition (Pearson Education, Harlow, 2014), 513 pp.
- <sup>4</sup>L. Eberhardt, *Off-shell Partition Functions in 3d Gravity*, (Apr. 28, 2022) arXiv:2204.09789 [hep-th], <http://arxiv.org/abs/2204.09789> (visited on 05/13/2025), pre-published.

## Todo Notes / Open Points